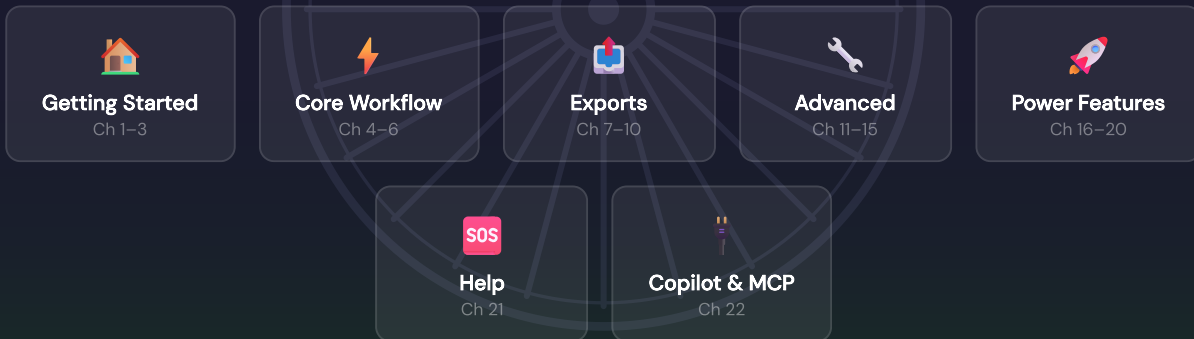


Complete Documentation · v12.20

Shuddhi QA

Shuddhi (शुद्धि) means "purification" in Sanskrit — clean test cases for clean code. AI-powered test case generation for enterprise QA teams.

[Jump to chapter](#)

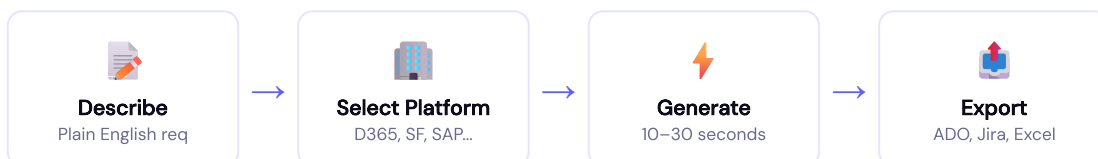
Overview & First Launch

What Shuddhi QA does and how to open it



What does Shuddhi QA do?

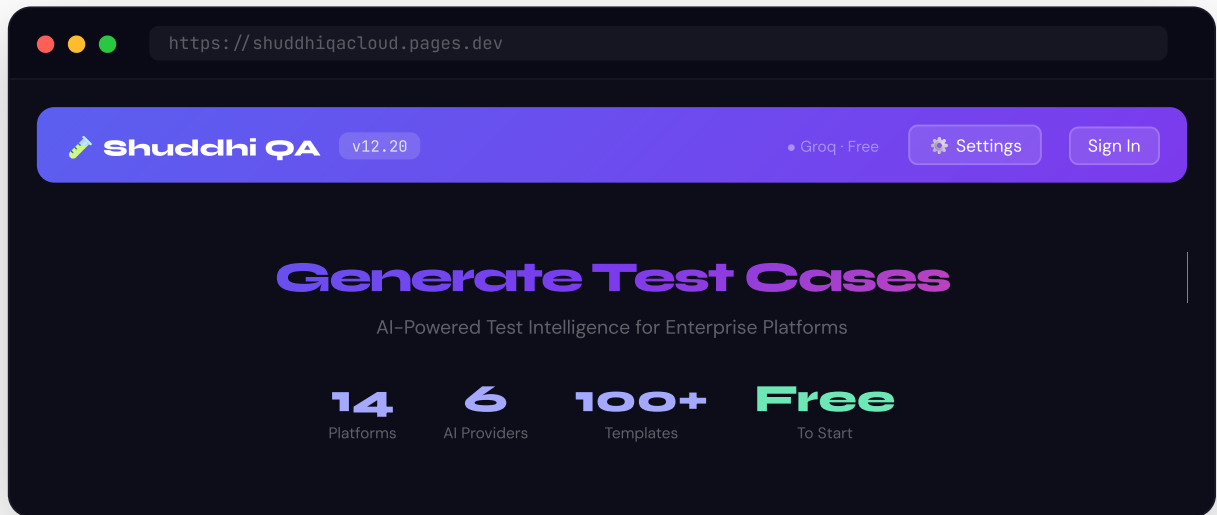
Shuddhi QA converts plain-English business requirements into production-ready test cases with steps, expected results, and quality scores — in under 30 seconds. No manual writing. No templates to fill.





Open the App

Open any browser and go to <https://shuddhiqacloud.pages.dev> — no installation, login, or credit card needed.



✓ **Free to use.** 3 generations anonymously · 10 with Google/Microsoft sign-in · Unlimited with your own API key.



Understand the Two-Panel Layout

The app is split into two panels. Left panel = your inputs (platform, module, requirement). Right panel = AI-generated output (test cases, export buttons).

The screenshot shows the Shuddhi QA application interface. The top navigation bar is purple with the Shuddhi QA logo, version v12.20, a settings gear icon, and a Sign In button. The main content area is split into two panels. The left panel, titled 'INPUT PANEL', contains three sections: 1. PLATFORM with buttons for D365 F&O, Salesforce, Web App, ServiceNow, SAP, and Workday. 2. MODULE with a text input field containing 'AP Invoice Processing'. 3. REQUIREMENT with a text area containing 'Vendor submits invoice. AP Clerk performs 3-way match (Invoice, PO, GR). If amount ≤ ₹50K auto-approved...'. Below these is a large purple button labeled 'Generate Test Cases' with a lightning bolt icon and the text 'Groq · Free (server-side)'. The right panel, titled 'OUTPUT PANEL', shows a summary bar with 'D365 F&O · AP Invoice · 12 TCs · Good 97%'. Below this is a table of test cases with columns for ID, description, and status. The table lists four test cases: TC001 (E2E: Successful 3-Way Match & Auto-Approval (≤₹50K) with 95% status), TC002 (E2E: Finance Manager Approval (> ₹50K) with 100% status), TC003 (E2E: 3-Way Match Failure — Qty Mismatch with 75% status), and TC004 (Validation: Invoice Amount Exceeds Tolerance with 100% status). At the bottom of the table, it says '+ 8 more test cases'.

ID	Description	Status
TC001	E2E: Successful 3-Way Match & Auto-Approval (≤₹50K)	95%
TC002	E2E: Finance Manager Approval (> ₹50K)	100%
TC003	E2E: 3-Way Match Failure — Qty Mismatch	75%
TC004	Validation: Invoice Amount Exceeds Tolerance	100%



Configure AI Provider

Choose and set up your AI provider

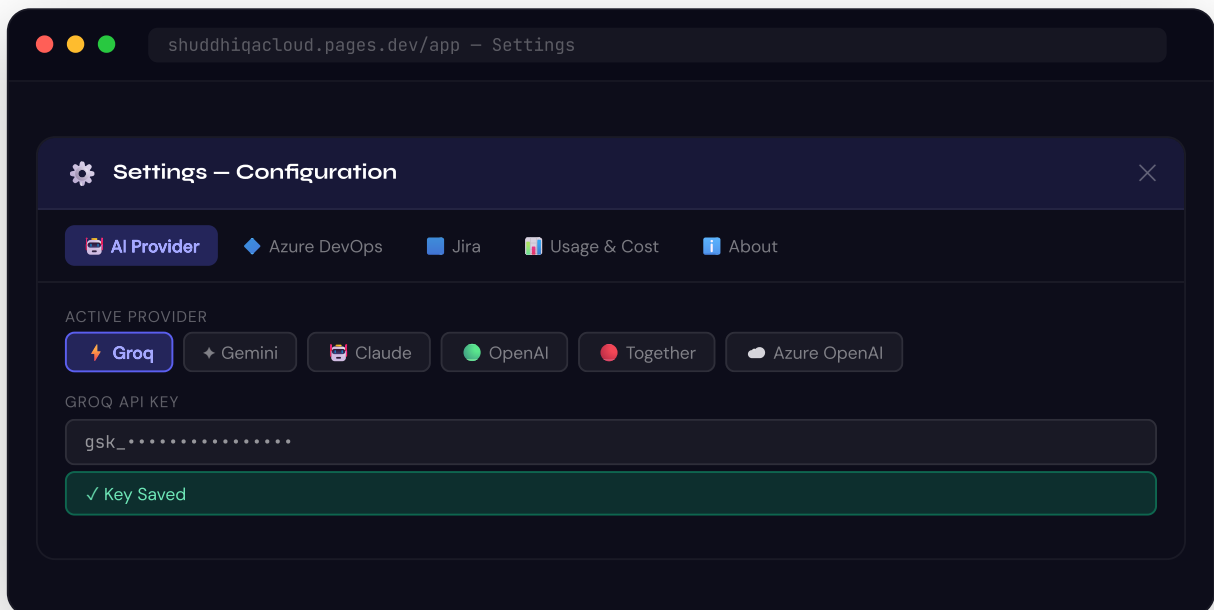
💡 **Skip for now.** The app uses a shared Groq server key for your first 3 free generations. Add your own key only when you want unlimited usage.

1 Compare the 6 AI Providers

Provider	Model	Cost / Run	Best For
⚡ Groq	Llama 3.3 70B	Free	Fastest responses, high volume
✦ Gemini	2.5 Flash	~\$0.001	Large requirements, best value
🤖 Claude	Sonnet 4	~\$0.015	Best quality, detailed steps
🟢 OpenAI	GPT-4o	~\$0.010	Complex multi-role flows
🔴 Together AI	Open source	~\$0.002	Cost-effective alternative
☁️ Azure OpenAI	GPT-4o (your deployment)	Your Azure rates	Enterprise · BYO — data residency in your region

2 Open Settings → AI Provider Tab

Click the ⚙️ **Settings** gear icon (top right) → click the **AI Provider** tab.





Get your API Key (Provider-specific)

Provider	Where to get your free key
 Groq (Free)	console.groq.com → API Keys → Create API Key
 Gemini	aistudio.google.com → Get API Key → Create in new project
 Claude	console.anthropic.com → API Keys → Create Key
 OpenAI	platform.openai.com → API Keys → Create new secret key
 Together AI	api.together.ai → Settings → API Keys
 Azure OpenAI	portal.azure.com → your Azure OpenAI resource → Keys and Endpoint (BYO — bring your own resource)



Privacy guarantee: Keys are stored in browser localStorage only. They never touch Shuddhi QA servers — requests go directly from your browser to the AI provider.



Azure OpenAI is different. Unlike the other 5 providers (one API key each), Azure needs **four** values. Open **Settings** → **AI Provider** → **Your Personal API Keys** → the expandable **"Azure OpenAI"** row and fill in: **Endpoint** (<https://<your-resource>.openai.azure.com>), **API Key**, **Deployment name** (e.g. gpt-4o-prod), and **API Version** (default 2024-12-01-preview). Then click **Test connection**. Generation runs in the region of *your own* Azure resource — ideal for enterprise data residency.

4

Quick provider switching (v12.20+)

If you configure 2 or more provider keys, you don't need to open Settings every time you want to switch between them. A small ⇌ icon appears next to the cost hint below the Generate button. Click the cost hint to open a popover showing all your configured providers with their costs at a glance.

How it behaves

- **0 or 1 provider configured:** cost hint is static (no ⇌ icon) — nothing to switch to
- **2+ providers configured:** ⇌ icon appears; click hint → popover opens
- **Active provider** is marked with a filled radio + "✓ active" tag
- **Costs shown right-aligned** in the popover — easy to scan free vs paid
- **Closes on** selecting a new provider, clicking outside, or pressing Esc



Why this exists: Most users pick one provider and stay there. But if you want to A/B test Claude vs Gemini on the same requirement, the popover saves you 3 clicks each switch. Settings remains the single source of truth — the popover is just a shortcut.



Platform & Module Setup

Select the right platform for domain-aware test cases

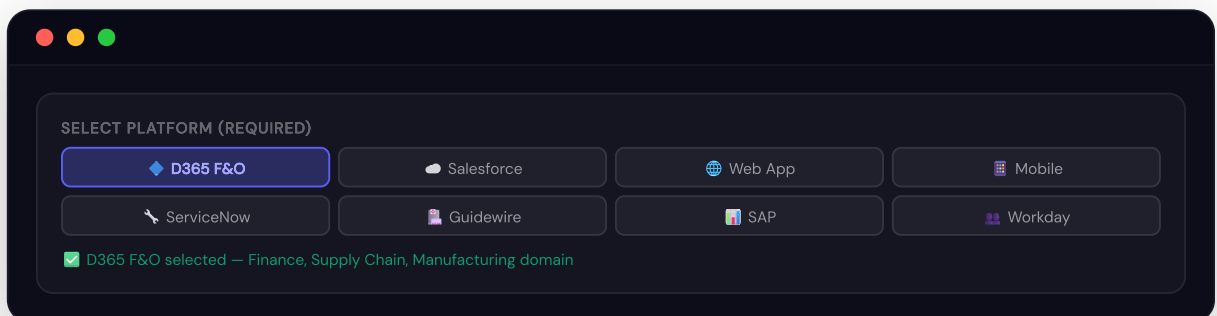


Both are required (v12.16+). Generation is blocked until you select a platform AND enter a module name. A red glow appears on the missing field.

1

Select Platform

Click one of the platform chips. The platform tells the AI which terminology, navigation paths, and field names to use.



2

Enter Module Name

Type the specific module or feature being tested. This gives the AI more context for accurate test case titles and navigation steps.



Good module names: "AP Invoice Processing" · "Vendor Payment" · "Opportunity Management" · "ITSM Incident" · "Leave Management" · "Purchase Order Approval"

3

Smart Detect – Auto Platform Detection

Paste your requirement first and Shuddhi QA automatically identifies the platform from the content. A green notification confirms the detected platform.

Keywords in text	Auto-detected Platform
Action Pane, infolog, legal entity, financial dimension	◆ D365 F&O
SOQL, LWC, Apex trigger, Opportunity, Account	☁ Salesforce
PolicyCenter, ClaimCenter, underwriting	📄 Guidewire
incident, ITSM, service catalog, ServiceNow	🔧 ServiceNow
S/4HANA, Fiori, T-code, SAP transaction	🏢 SAP



Write Your Requirement

Writing requirements that generate the best test cases

1

Paste or type your requirement

Click the requirement textarea and paste your user story, BRD excerpt, or process description. Maximum 6,000 characters. The counter updates as you type.

DESCRIBE YOUR REQUIREMENT (REQUIRED)

Vendor submits invoice via portal. AP Clerk performs 3-way match (Invoice, PO, Goods Receipt). If amount $\leq$ ₹50,000, auto-approved. If $>$ ₹50,000, Finance Manager must approve within 2 business days. On approval, payment is scheduled to vendor bank account. On rejection, vendor receives email with reason code.

347 / 6,000

2

What makes a good requirement

✓ Include	✗ Avoid
User roles (AP Clerk, Finance Manager)	Generic "test the invoice process"
Thresholds ($\leq$ ₹50,000 / $>$ ₹50,000)	No boundary values
Expected outcomes (payment scheduled)	No expected results stated
Error paths (rejection $\rightarrow$ email)	Only happy path described
200–500 characters minimum	Single sentence



Choose Output Type

Type	What gets generated	When to use
E2E Scenarios	Full end-to-end flows	Sprint testing, regression
Smoke Tests	Critical path only	After deployments
Field Validation	Input validations, boundaries	Form testing
Integration Tests	System integration points	API/middleware
Negative Tests	Error paths, invalid inputs	Security, robustness
Full Artifact	All 5 types combined	Comprehensive coverage



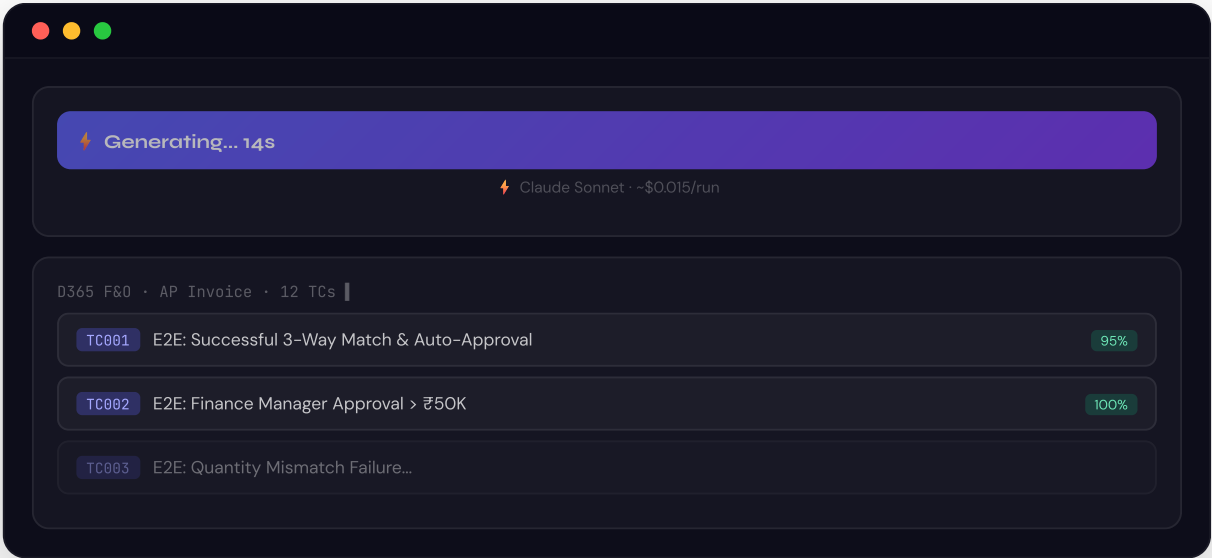
Generate Test Cases

Trigger generation and interpret the output



Click Generate or press Ctrl+Enter

Click ⚡ **Generate Test Cases** or press **Ctrl** + **Enter**. Test cases stream into the right panel in real-time. Takes 10–30 seconds depending on provider.





Understand Quality Scores

Score	Badge Color	Meaning	Action
90–100%	● Good	Complete, accurate, well-structured	Export as-is
70–89%	● Fair	Usable, minor edits may help	Quick review
<70%	● Needs Review	Limited context available	Edit or regenerate



Pre-generation Validation Errors

Error Message	Fix
"No AI provider configured"	Settings → AI Provider → add Groq key (free)
"Please select a platform"	Click a platform chip (D365, Salesforce, etc.)
"Please select a module or enter a custom module name"	Type module name in the module field
"Please describe the business flow"	Type or paste requirement in textarea



Review & Edit Test Cases

Inline editing, select-all, and quality review



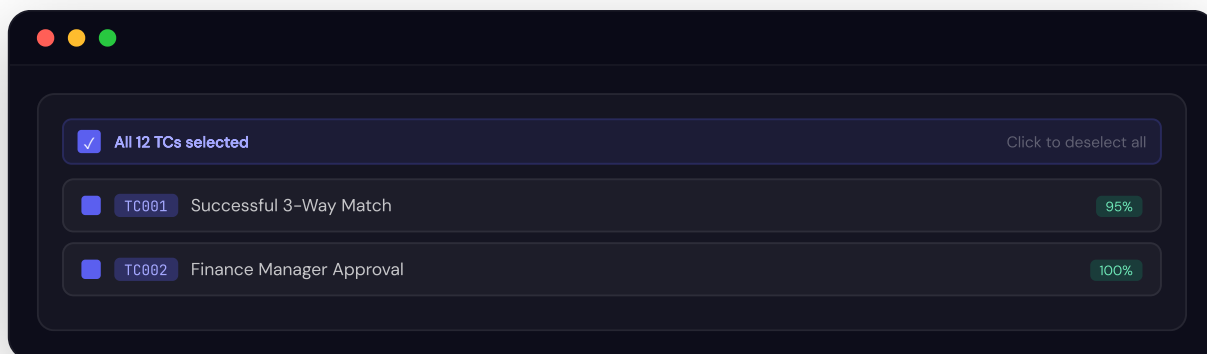
Enter Inline Edit Mode

Click  **Edit** in the export toolbar. Every cell in the table becomes clickable and editable. Changes reflect immediately in all export formats.



Select / Deselect All (v12.15+)

Use the header checkbox to select or deselect all test cases. Only selected TCs are included in exports.



 **Toggle logic:** None → click → All selected. All → click → None. Some → indeterminate dash → click → All selected.



Export to Excel, PDF & Markdown

Download test cases for stakeholder review



Excel (.xlsx)

Click  **Excel**. Downloads .xlsx with separate sheets per TC type. Only selected TCs included.



PDF

Click  **PDF**. Formatted document with all test cases and steps — ideal for sign-off packages.



Markdown

Click  **Markdown**. Copies to clipboard — paste into Confluence, Notion, or any Markdown editor.



Push to Azure DevOps

One-click push to ADO Test Plans

1

Create ADO Personal Access Token (PAT)

Steps to create ADO PAT

- Go to dev.azure.com → your organization
- Click your profile icon (top right) → Personal Access Tokens
- Click + New Token
- Name: ShuddhiQA-PAT · Expiration: 90 days
- Scopes: Test Plans (Read & Write) + Work Items (Read & Write)
- Click Create → Copy the token immediately (shown once only)

2

Configure in Settings → Azure DevOps Tab

Settings — Azure DevOps

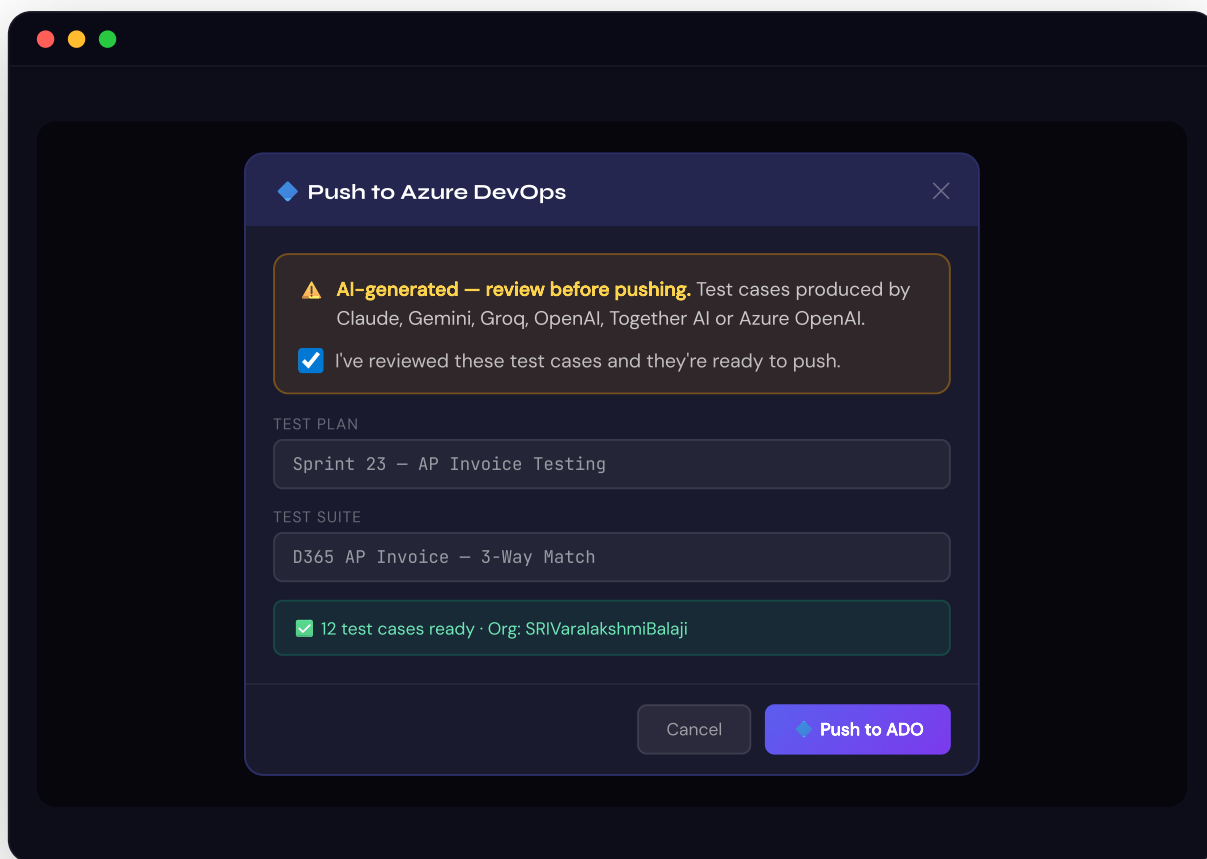
ORGANIZATION URL

PERSONAL ACCESS TOKEN



Push Test Cases to ADO

Click  **ADO** in the export toolbar → Select Test Plan → **tick the AI-review acknowledgement** → Click Push. The Push button stays disabled (45% opacity) until the checkbox is ticked — see [Ch 19](#) for why this gate exists.



- ✓ **What gets created:** Test Plan · Test Suite per module · Test Cases with full steps · Expected results · State: Design



Export to Jira Zephyr

CSV for Zephyr Squad or Scale import



Get Atlassian API Token

Steps to get API token

- Go to id.atlassian.com → Security → API tokens
- Click Create API token · Name: ShuddhiQA
- Click Create → Copy the token



Configure in Settings → Jira Tab

Settings → Jira tab → Enter token + email + domain (e.g. yourcompany.atlassian.net without <https://>).



Export & Import

After generating: click **More ▾** →  **Jira Zephyr**. CSV downloads with all 10 Zephyr headers pre-populated. In Jira: Zephyr → Test Cases → Import → Upload CSV.



You'll see a toast on download: "Jira CSV downloaded — review in Excel before uploading to Jira." This is the AI-review reminder — see [Ch 19](#) for the full explanation of why this gate exists.



Other Export Formats

Gherkin, Playwright, TestRail — via **More ▾** dropdown

Format	Use Case	Access
 Gherkin	BDD — Cucumber, SpecFlow, Behave	More ▾ → Gherkin
 Playwright	TypeScript automation skeleton	More ▾ → Playwright
 TestRail	CSV import for TestRail	More ▾ → TestRail
 Postman	API test collection skeleton	More ▾ → Postman



More ▾ button in the export toolbar reveals additional formats. Click once to expand, once more to close.



Iterative Refinement

Regenerate individual test cases without affecting others

1

Hover → click on any test case row

Move your mouse over any test case row. A regenerate icon appears on the right. Click it to open the refinement modal.

2

Add context and regenerate

 Only that one TC is updated. The rest of the suite stays intact.



Templates & Smart Detect

100+ pre-built templates and auto platform detection



Using Templates

Visit shuddhiqacloud.pages.dev/templates or click Templates in the nav. Browse 100+ pre-built requirement templates. Click any to load it into the requirement field instantly.



Smart Detect on Paste

When you paste 100+ characters, the AI analyzes your text and suggests the platform. A notification appears confirming detection.



Smart Detect runs automatically on every paste/input event. No button to click — it fires in the background.



Sign In & Privacy

Optional sign-in for cross-device sync

Feature	Anonymous	Signed In
Free generations	3 per session	10 per session
Preference sync	Browser only	✓ All devices
History	✗	✓ Last 30 days
GDPR delete	N/A	✓ Full deletion



Only email + name accessed. No Gmail, Drive, Calendar access. Standard OAuth 2.0 with minimal scopes. Keys are never synced to Firebase — browser localStorage only.



Shortcuts & Command Palette

Speed up your workflow

Action	Shortcut
Generate test cases	Ctrl + Enter
Open command palette	Ctrl + K
Toggle dark/light theme	Ctrl + Shift + T
Close modal / palette	Esc
Open Settings	Ctrl + ,
Copy Markdown	Ctrl + C



Command Palette (Ctrl+K) — search across all 19 commands. Type "dark", "export", "templates", "settings" — any command appears instantly.



Settings Reference

All 5 settings tabs explained

Tab	What you configure
AI Provider	Select active provider. Paste API keys for Groq, Gemini, Claude, OpenAI, Together AI. Azure OpenAI (BYO) takes Endpoint + Key + Deployment + API Version.
Azure DevOps	Org URL + PAT. Test Connection button. Validates PAT format before saving.
Jira	Atlassian API token + email + Jira domain. Test Connection button.
Usage & Cost	Generation count tracker. Google Cloud Billing sync (OAuth). Token usage estimates.
About	Version number, active provider, platform diagnostics, changelog history.



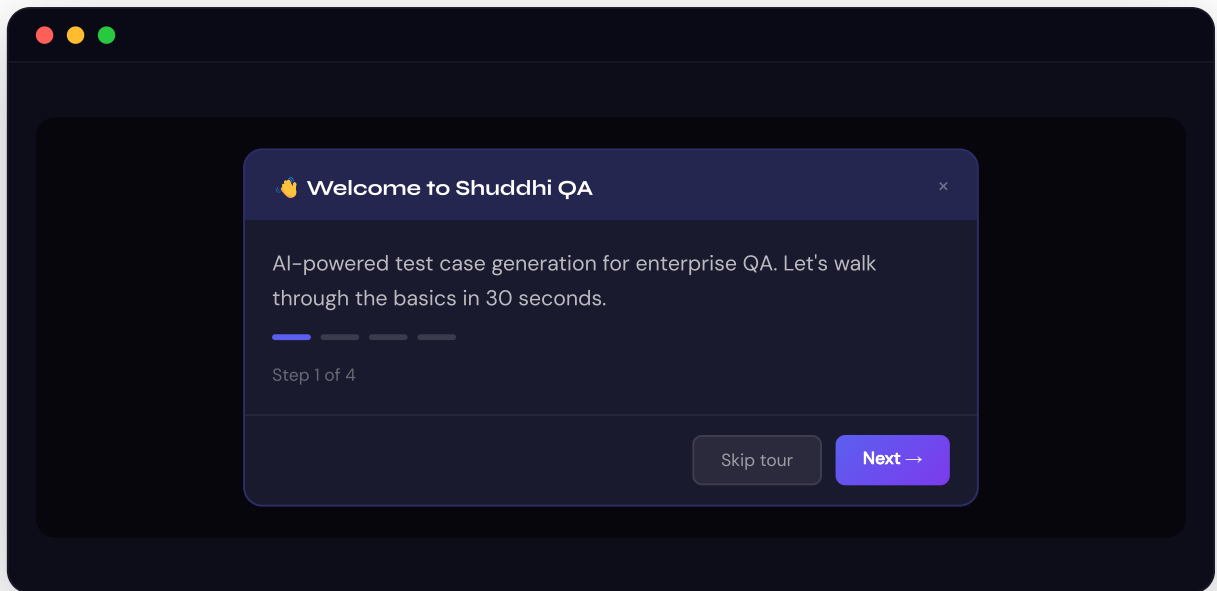
Welcome Tour & First-Time Experience

The 4-step walkthrough that runs on first visit

1

Auto-shows on your first visit

The Welcome modal auto-launches the first time you load `shuddhiqacloud.pages.dev/app` — even before signing in. It's tracked via a single browser flag (`shuddhi-first-run-done`) in `localStorage`, so it appears exactly once per device.



2

The 4 tour steps

Step	What it teaches
1. Welcome	Intro and the "what is Shuddhi QA" pitch
2. Select a Platform	Highlights the platform chips on the left panel
3. Describe Requirement	Points to the textarea + character counter
4. Generate & Export	Shows the Generate button + export toolbar with ADO/Jira

💡 Each step highlights the relevant UI region with a pulse animation. You can **Skip tour** anytime — the flag still flips so the modal doesn't reappear next visit.



How to re-show the tour

The tour is one-shot per device, but you can manually replay it:

To re-trigger the welcome modal

- Open DevTools (F12) → Application tab → Local Storage → shuddhiqacloud.pages.dev
- Delete the `shuddhi-first-run-done` key
- Refresh the page — the tour appears again



If you're writing E2E tests: Playwright scripts can set the `shuddhi-first-run-done` flag via `addInitScript` before each test, so the welcome modal doesn't block other UI interactions. Skip this if you're not running automated tests.



Domain Intelligence

Vertical-aware test generation with business-risk tagging



Distinct from Platform. Platform = the tech stack (D365, Salesforce). Domain = the business vertical (Insurance, Banking, Healthcare). You can mix and match — e.g. Salesforce + Insurance domain produces underwriting and policy-binding test cases.

1

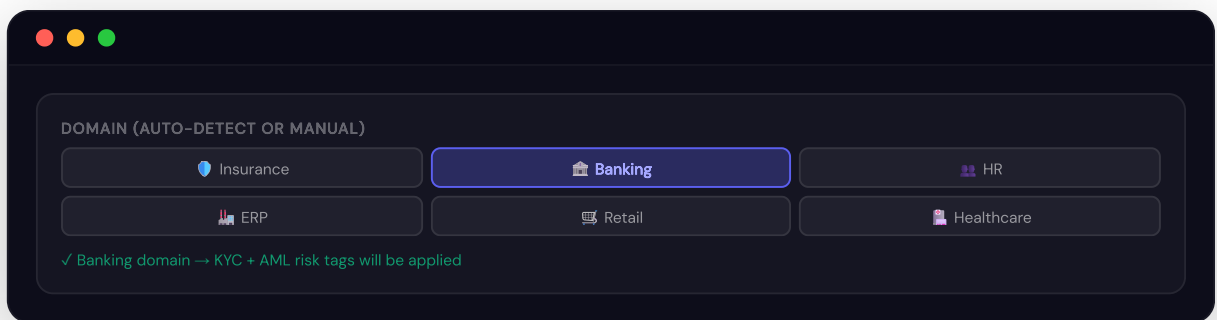
6 Supported Domains

Domain	Terminology added	Risk patterns
 Insurance	Policy, claim, premium, underwriting, FNOL (First Notice of Loss)	Regulatory compliance, fraud, claims leakage
 Banking	KYC (Know Your Customer), AML (Anti-Money Laundering), ledger, settlement, reconciliation	Compliance breach, double-debit, audit trail
 HR	Onboarding, payroll, leave, performance review	PII (Personally Identifiable Info) exposure, payroll calc errors, GDPR
 ERP	3-way match, MRP (Material Requirements Planning), BOM (Bill of Materials), ledger posting	Stockout, posting reversal, period-close errors
 Retail	Loyalty, returns, omnichannel, inventory	Oversell, price mismatch, fulfillment delay
 Healthcare	EMR (Electronic Medical Records), HL7 (data exchange standard), claims, prior auth, HIPAA	PHI (Protected Health Info) exposure, prior-auth denial, coding errors

2

Selecting a domain

Click the **Domain** selector above the requirement textarea. Default is "Auto-detect" — Shuddhi QA picks the domain from your requirement keywords (e.g. "policy", "KYC", "HL7" → triggers the matching vertical).



The screenshot shows a dark-themed window titled "DOMAIN (AUTO-DETECT OR MANUAL)". It contains six buttons arranged in a 2x3 grid, each with an icon and a label: Insurance (shield icon), Banking (bank icon), HR (people icon), ERP (factory icon), Retail (shopping cart icon), and Healthcare (pill icon). The "Banking" button is highlighted with a blue border. Below the buttons, a green status message reads: "✓ Banking domain → KYC + AML risk tags will be applied".



What changes in the output

Without domain	With Insurance domain
TC: "Verify form submission"	TC: "Verify FNOL submission triggers claim creation in PolicyCenter"
No risk tag	Tag: ● Regulatory
Generic field names	Specific terms (premium, deductible, claimant)
Generic personas	Underwriter, claims adjuster, broker



Multi-Language Output

Generate test cases in 5 languages — 4 more coming soon



Currently supported (Tier 1 — full QA translation)

Language	Code	Best for
GB English	en	Default — global teams, ADO/Jira standard
DE Deutsch (German)	de	DACH region, automotive/manufacturing
ES Español (Spanish)	es	LATAM, Spain operations
FR Français (French)	fr	France, Quebec, francophone Africa
BR Português (BR)	pt-BR	Brazilian enterprises



Coming Soon (Tier 2): JP 日本語 (Japanese) · NL Nederlands (Dutch) · IT Italiano (Italian) · PL Polski (Polish). These appear in the dropdown but are disabled until full QA testing completes.



How to select

Choose your output language from the **Language** dropdown in the left panel, just above the Generate button. Selection persists across sessions and syncs to your account if signed in.



You can write requirements in English and request output in any other language. Domain terms (e.g. "underwriting", "3-way match") are translated to their localized equivalent rather than left in English.



What gets translated

Element	Translated?	Notes
Test case titles	✓ Yes	Localized including domain terminology
Step actions & expected results	✓ Yes	Full localization
UI button labels referenced in steps	✓ Yes	e.g. "Speichern" instead of "Save" for German
Technical identifiers (TC001, field names)	✗ Kept in English	For traceability with ADO/Jira
Excel/PDF column headers	✓ Yes	Headers match selected language



ADO/Jira push: Test case content uses your selected language but the platform's work-item fields (State, Priority, Assigned To) stay in ADO/Jira's UI language.



AI-Review Gate

Mandatory review before pushing to ADO or Jira



Added because AI can hallucinate. Before any test case reaches your production work items, you must explicitly confirm you've reviewed them. The Push button stays disabled until you tick the acknowledgement.

1

ADO push – review checkbox

When you click  **ADO**, the push modal shows an amber warning banner above the Test Plan field with the text *"AI-generated – review before pushing"* and a required acknowledgement checkbox:

☒ **"I've reviewed these test cases and they're ready to push."** Until you tick this, the Push button stays disabled at 45% opacity – no accidental clicks possible.



See [Ch 8 step 3](#) for the modal mockup with the checkbox in context.

2

Jira Zephyr – review reminder on download

Jira uses CSV import (no API push), so the review gate fires as a toast notification when the CSV is downloaded.



Toast text: "Jira CSV downloaded – review in Excel before uploading to Jira"



Workflow: download CSV → open in Excel → cross-check each test case → only then upload to Jira via Zephyr → Test Cases → Import.

3

Why this matters

Without the gate	With the gate
50+ hallucinated test cases in your sprint	You catch them before sprint planning
QA team blames "the AI tool" for bad TCs	Accountability stays with the human reviewer
Production incidents from untested edge cases	Edge cases verified before they go live



Internal audit trail: The checkbox tick is logged in your Firestore session record (if signed in) – useful for SOC2 / ISO27001 process evidence.



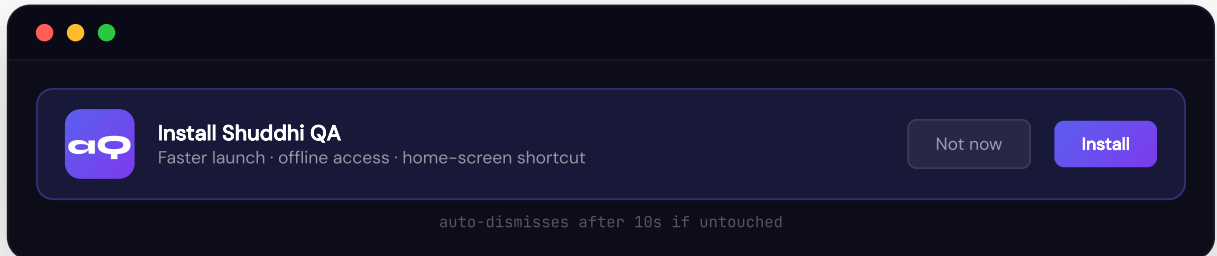
PWA Install & Offline Mode

Install Shuddhi QA as an app on desktop or mobile



The Install banner

On the home page (shuddhiqacloud.pages.dev/), an "Install Shuddhi QA" banner appears at the bottom of the screen. Click **Install** to add it to your OS as a standalone app — like installing any native app.



2

What you get after installing



Faster launch

App opens in its own window, no browser chrome. ~40% faster cold start than a fresh tab.



Offline access

App shell + UI loads without network. Generation still needs a connection, but you can review prior TCs offline.



Home-screen icon

aQ icon on indigo background — same on Windows Start, macOS Dock, iOS/Android home screen.



Push-ready

Foundation for future browser push notifications (e.g. "Your generation is ready").

3

Stuck on an old version?

If the app feels stale after a recent update, force a fresh fetch with **Ctrl+Shift+R** (hard refresh). That clears the cached version and pulls the latest from the server. For most users, this is all you ever need to do.



For developers / SDETs: The PWA uses a service worker (sw.js) that caches the app shell — app.html, index.html, manifest.json, icons, fonts. Only network-bound calls are /api/claude (generation), /api/ado (push), and Firebase Auth flows. To inspect or force-update: DevTools → Application → Service Workers → "Update on reload".



Troubleshooting

Common issues and how to fix them

Issue	Cause	Fix
Generate button not working	No platform or module selected	Select a platform chip → enter module name
"No AI provider configured"	No API key set and server key unavailable	Settings → AI Provider → add Groq key (free)
ADO push fails with 401	PAT expired or wrong scope	Create new PAT with Test Plans (Read & Write) scope
"PAT invalid or expired"	Bogus or expired ADO PAT	Generate a new PAT in ADO user settings
Firebase sign-in fails (network error)	Browser extension blocking requests	Try Incognito mode or disable extensions
Generated TCs are too generic	Requirement too brief	Add roles, thresholds, and expected outcomes to requirement
Character limit exceeded	Requirement over 6,000 chars	App auto-trims with notification — reduce if needed
More ▾ dropdown not opening	Old browser cache (pre-v12.4)	Hard refresh Ctrl+Shift+R
Cost hint not showing	Provider not initialized yet	Wait 2–3 seconds after page load
Jira CSV import errors	Wrong domain format in Settings	Use yourcompany.atlassian.net (no https://, no trailing slash)



Help page: Visit shuddhiqacloud.pages.dev/help for detailed setup guides for each AI provider, ADO, and Jira configuration steps.



Still stuck? Email ramya9.b@gmail.com with a screenshot and the browser console output (F12 → Console tab).



Microsoft Copilot & MCP Integration

Use Shuddhi QA from inside Copilot, Claude Desktop, Cursor, and any MCP-capable AI client



Live now in v12.20. Shuddhi QA is exposed as a Model Context Protocol (MCP) server. Add it to any MCP-capable AI client in 30 seconds. Five tools are available: `generateTestCases`, `fetchJiraIssue`, `pushToADO`,

1

What is Model Context Protocol (MCP)?

MCP is an open protocol released by Anthropic in 2024 and adopted by Microsoft, OpenAI, and others. It lets AI assistants discover and use external tools/resources through a standard JSON-RPC 2.0 interface. Shuddhi QA implements an MCP server so any compatible AI client can invoke its test case generation, Jira fetch, and Azure DevOps push capabilities directly from chat — without leaving the AI interface.

2

MCP server endpoint

The Shuddhi QA MCP server is hosted on Cloudflare Workers at:

Field	Value
Endpoint URL	https://shuddhiqa-mcp.ramya9-b.workers.dev/mcp
Protocol	JSON-RPC 2.0 over HTTP POST (Streamable HTTP transport)
MCP version	2025-06-18
Server version	0.1.0
Authentication	None (v0.1) — anonymous + Cloudflare DDoS protection
Health probe	GET /healthz returns { "ok": true }
Server info	GET / returns server name, version, protocol, and tool names



Five tools exposed

Tool	Purpose	Backend
generateTestCases	Generate production-grade QA test cases for 14 enterprise platforms with real navigation paths, T-codes, table names, validation rule names, and platform-specific error messages.	Proxies to /api/claude
fetchJiraIssue	Pull a Jira ticket's summary, description, acceptance criteria, status, and assignee. Caller supplies base URL, email, and API token.	Proxies to /api/jira
pushToADO	Push generated test cases to an Azure DevOps Test Plan as Test Case work items. Optionally adds to a specific suite.	Proxies to /api/ado
listTemplates	Browse a curated subset of the 110-template library, filtered by platform and business domain.	In-memory (bundled)
detectPlatform	Identify the enterprise platform, business domain, and module from any requirement text. Returns confidence score and supporting semantic evidence.	Proxies to /api/detect



Setup — Microsoft 365 Copilot (via Copilot Studio)

Prerequisite: active Microsoft 365 Copilot license; tenant admin allows custom Copilot Studio agents with MCP connectors.

1. Open **copilotstudio.microsoft.com** with your work account.
2. Click **Create** → **New agent** → choose **Declarative Agent**.
3. Name it **Shuddhi QA**; paste the description from the homepage.
4. In **Knowledge / Actions**, click **Add** → **MCP server**.
5. Paste the URL: `https://shuddhiqa-mcp.ramya9-b.workers.dev/mcp`
6. Authentication: **None** (v0.1).
7. Click **Test** — Copilot Studio should discover all 5 tools.
8. **Publish**, then invoke from any Copilot surface: `@Shuddhi-QA generate D365 vendor invoice test cases.`

5

Setup — Claude Desktop

Edit your Claude Desktop config file. On Windows: %APPDATA%\Claude\claude_desktop_config.json. On macOS: ~/Library/Application Support/Claude/claude_desktop_config.json. Add:

```
{
  "mcpServers": {
    "shuddhi-qa": {
      "transport": {
        "type": "streamable-http",
        "url": "https://shuddhiqa-mcp.ramya9-b.workers.dev/mcp"
      }
    }
  }
}
```

Restart Claude Desktop. The Shuddhi QA tool surface appears in the attachments / tool list.

6

Setup — Cursor, Continue, and other MCP clients

In Cursor: **Settings** → **Cursor Settings** → **MCP Servers** → **Add**. Type: **Streamable HTTP**. URL: `https://shuddhiqa-mcp.ramya9-b.workers.dev/mcp`. Auth: **None**.

In Continue (VS Code / JetBrains): edit `~/.continue/config.json` and add the same `mcpServers` block as Claude Desktop.

For any other MCP-capable client, paste the URL into whatever "Add MCP server" form they expose.

7

Authentication for Jira and Azure DevOps

When you invoke `fetchJiraIssue` or `pushToADO` from your AI client, you supply credentials per-call. They are passed through the MCP server to the target service and discarded after the request completes — never logged, never stored.

Tool	Required credentials
<code>fetchJiraIssue</code>	Jira base URL, account email, API token (from id.atlassian.com)
<code>pushToADO</code>	Azure DevOps organization, project, test plan ID, PAT with Test Plans (Read & Write) scope



Example Copilot conversation

User: @Shuddhi-QA pull Jira ticket PROJ-1234 and tell me what platform it targets

Copilot: [calls fetchJiraIssue → detectPlatform]
"PROJ-1234 is a D365 F&O vendor invoice requirement, 92% confidence.
The ticket references workflow approval over \$10K and three-way matching."

User: Generate test cases for it

Copilot: [calls generateTestCases]
"Here are 40 D365 vendor invoice test cases:
TC001 – Navigate to Accounts Payable > Invoices > Vendor invoices...
TC002 – Verify VendInvoiceInfoTable.WorkflowStatusEnum = InReview..."

User: Push these to ADO Test Plan 567

Copilot: [calls pushToADO]
"Pushed 40 test cases to Plan 567 in suite 1024.
Work item URLs: dev.azure.com/contoso/RetailPlatform/_workitems/..."



Privacy and data flow

The MCP server is stateless. No database. No persistent storage of customer content. It operates as a thin protocol adapter between MCP clients and the Shuddhi QA backend.

- **We store:** Operational metrics (request counts, error rates, tool-name frequencies) for 90 days, then aggregated to monthly counts for 12 months.
- **We never store:** Requirement text, Jira ticket content, generated test cases, or any caller-supplied credentials.
- **We forward:** Requirement text to your selected AI provider (Anthropic, Google, Groq, OpenAI, Together AI, or your own Azure OpenAI resource). Jira credentials to your Jira instance. ADO credentials to dev.azure.com.

Full privacy disclosures: shuddhiqacloud.pages.dev/privacy → "Microsoft Copilot & MCP Integration" section.



Microsoft AppSource listing (planned)

For globally discoverable distribution to every Microsoft 365 Copilot tenant, Shuddhi QA is in active preparation for Microsoft AppSource submission. Once listed, tenant admins will be able to install Shuddhi QA tenant-wide from the Microsoft 365 admin center in a single click — no manual Copilot Studio configuration required.



Current path: Until AppSource listing goes live, the manual Copilot Studio setup above (step 4) is the canonical install path. Once AppSource certification is complete, the install URL will be added to the homepage and this section will be updated with the AppSource link.



Roadmap – v0.2 and beyond

- Authentication / API keys for caller identification and per-tenant quota
- Full 110-template enumeration via `listTemplates` (v0.1 surfaces 28 curated)
- Additional tools: `pushToJira`, `exportPlaywright`, `scoreTestCases`, `fetchAdoWorkItem`
- Azure OpenAI as a 6th AI provider for enterprises with data residency requirements (✓ shipped in v12.21)
- Microsoft Entra ID authentication for organizational governance



Open source: Full MCP server source code at github.com/ramya9b/shuddhiqacloud. Self-host to your own Cloudflare Workers account if data residency or compliance requires it.



Appendix A – Document Index

Other guides and reference documents



This is internal reference for the documentation team. End users only need this Documentation plus the in-app help pages.



Step-by-Step User Guide

15-chapter interactive HTML guide with UI mockups covering every feature from first launch to enterprise integrations. Sticky sidebar navigation.

</guides/step-by-step.html>



Feature Overview HTML

Interactive HTML version of the feature guide with sticky nav, hover effects, provider comparison, export cards, test coverage grid.

</guides/features.html>



Feature Overview Presentation (internal)

12-slide PPTX deck with indigo/dark premium theme. For stakeholder presentations. Available on request.

[ShuddhiQA_Features.pptx](#)



Word Document User Guide – v7.9 era (archived)

35-page .docx historical guide. 13 sections with embedded screenshots. Preserved for reference; superseded by this Documentation.

[ShuddhiQA_v79_UserGuide.docx](#)



Deployment Packages (v10.1 → v11.3) — internal

Versioned Cloudflare Pages deployment zips containing complete app source. Internal release artifacts.

ShuddhiQA-v10.x through v11.3.zip



The current live version is **v12.20** at shuddhiqacloud.pages.dev. For access to internal/archived documents, contact ramya9.b@gmail.com.



Appendix B — Version History

Key milestones from v7.x to v12.21

v12.21

Azure OpenAI added as a 6th AI provider (Enterprise · BYO). Configured under Settings → AI Provider with four values — Endpoint, API Key, Deployment name, and API Version (default 2024-12-01-preview) — plus a Test connection button. Generation runs in the region of the user's own Azure resource for data residency.

v12.20

Format-aware generation (single-section formats now respect the dropdown). Demo videos on homepage. Honest 5-language marketing (was 9). New feature cards: AI-Review Gate, Domain Intelligence, Quality Scoring, PWA. FAQ expanded to 17 questions. Export format fixes (UTF-8 BOM, metadata-row filter, phantom-step removal across Playwright/Gherkin/TestRail/Jira/Postman/Coverage). Quick-switch cost popover appears when 2+ providers are configured — no need to open Settings every switch. Master Guide renamed to Documentation.

v12.19

Lazy marked.js, Firebase IDB-sniff lazy load, GIS dedupe — ~250KB saving on first load. Performance improvements.

v12.18

#genBtnCostHint — visible cost element below Generate button showing active provider and cost per run.

v12.17

No AI provider configured check added. Unconditional ProviderManager gate replaces silent __statusCheckFailed bypass.

v12.16

Platform and module selection made mandatory. Red glow highlight on validation failure. Cost hint below Generate button.

v12.15

Select-all toggle fixed (None→All, All→None, Some→Indeterminate). 11 browser scenarios verified.

v12.14

help.html hero pill version label updated. 25 missing CSS classes added to help.html — page was unstyled raw HTML.

v12.13

Critical fix: /terms hero and /help hero + 5 sections were invisible due to dead .fade-up opacity:0 CSS. 8 elements restored.

v12.12

privacy.html dead fade-up class cleanup. 2 elements unblocked.

v12.11

privacy.html hero eyebrow fade-up fix. Dead CSS rule removed.

v12.10

Google Sign-In always shows consent screen (prompt:consent). Firebase Web client redirect URI added.

● v12.9

GDPR deleteDoc Firestore fix — silent skip resolved. deleteDoc added to Firebase module imports.

● v12.8

_USER_KEY_LS bug fixed — OpenAI + Together AI user keys now reach generation pipeline.

● v12.7

Firebase SDK compat→modular migration. ORB blocking resolved. Version bumped to 9.23.0.

● v12.6

_initAutoSmartDetect called. #flowCharCount span added. wrangler.toml [assets] removed.

● v12.5

/api/ado PAT validation hardened. Bogus PATs rejected with clear error.

● v12.4

toggleMoreExports/closeMoreExports added. 5 aria-labels added. errText ReferenceError fixed.

● v11.x

Freemium gate, Firebase Auth, ADO/Jira proxy rewrites (Vercel→Cloudflare), lazy loading, conversation history.

● v10.x

Initial Cloudflare Pages deployment. Core generation pipeline. Provider manager. Export suite.

● v7.x–v9.x

Original Vercel → Cloudflare migration. Free trial gate. LinkedIn unlock. ADO/Jira integration. Gold/indigo brand.